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11 December 1987
SUPERSEDING
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17 July 1978

MILITARY SPECIFICATION

PUMP, PRESSURE, OIL FREE AIR
TYPE MA-3

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable 115-volt alternating current, electric motor drive, oil free air, pressure pump unit, designated Type MA-3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-E-489	Enamel, Alkyd, Glass (For Exterior And Interior Surfaces)
TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-C-5541	Chemical Conversion Coatings On Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4310
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MIL-M-7969	Motor, Alternating Current, 400-Cycle, 115/200-Volt System, Aircraft, General Specification For
MIL-A-8625	Anodic Coating, For Aluminum And Aluminum Alloys
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-889	Dissmilar Metals
MS3102	Connector, Receptacle, Electric, Box Mounted
MS33543	Criteria Temperature And Altitude Range Self Cooled Electric Equipment
MS33656	Fitting End, Standard Dimensions For Flared Tube Connection And Gasket Seal

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Applicaton for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.2 Components. The electric motor driven, oil free air pressure pump unit shall consist of a pump and an electric motor, built as an integral unit.

3.3 Materials. Materials and processes used in the manufacture of oil free air pressure pump units shall be of high quality and suitable for the purpose intended.

3.3.1 Metals.

3.3.1.1 Dissimilar. The use of dissimilar metals shall be avoided whenever practicable, or if they must be used, they shall be treated to give satisfactory protection in service and meet the other requirements stated herein. Dissimilar metals are defined by MIL-STD-889.

3.3.1.2 Castings. Castings shall be of uniform quality and condition free from cracks, excessive blow holes and porosity, and other injurious defects.

3.3.2 Protective treatment. When materials are used in the construction of air pressure pump units that are subject to corrosive conditions likely to occur during service usage, they shall be protected against corrosion resulting from these conditions in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions, shall be avoided.

3.3.3 Selection of materials. Specifications and standards for all materials, parts and government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.3.1 Standard parts. Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, practicable, on the contractor's drawings.

3.3.3.2 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.4 Design. The pump unit shall be designed to operate primarily as a pressure pump for maintaining sea level pressure in low capacity systems at high altitudes. The pump shall deliver oil free air at the outlet.

3.5 Environmental.

3.5.1 Temperature. The pump unit shall be capable of starting and operating satisfactorily at temperatures ranging from minus 65 to plus 160°F at altitudes varying from zero to 50,000 feet in accordance with curve II of MS33543.

3.5.2 Environmental inspections. The pump unit shall operate satisfactorily, as defined herein, when subjected to the conditions specified in 4.4.4.

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3.6 Rated capacity. The capacity of the unit shall be at least that shown in figure 1.

3.7 Mounting position. The pump unit shall operate satisfactorily in any mounting position without causing damage or shortening the life of the equipment.

3.8 Alternating current electric motors. Alternating current electric motors shall be in accordance with Specification MIL-M-7969.

3.9 Electric motor rotor.

3.9.1 Squirrel cage rotors. Squirrel cage rotors shall be used in all motors, except as noted herein. Where the application will be better served with a different type of rotor, such rotor may be used, provided that such rotor does not require the use of slip rings or brushes.

3.9.2 Rotor bars. The number of rotor bars shall be sufficient to permit the motor to develop the full required starting torque in any position of the rotor.

3.10 Check valve. A check valve shall be incorporated within the pump unit and placed in the outlet port, to prevent any leakage of air back into the pump unit when it is not running. The valve shall show no leakage back into the pump unit when a pressure differential of 25 inches of mercury is applied. The outlet of the check valve shall be in accordance with MS33656-5, Style G, for attachment of .313 inch O.D. tubing.

3.11 Electrical properties.

3.11.1 The electric motor, built integrally with the pump, shall operate continuously for 15 minutes out of every hour for 900 hours and 30 minutes out of ever hour for 100 hours.

3.11.2 Power. The power consumption under any condition, as required by the Capacity or Calibration test, shall be no greater than 750 watts at rated voltage. The power consumption for cold starting shall be not more than 6 amperes at each start; draw no more than 4.5 amperes, 15 seconds after starting; power required shall not exceed 350 watts after one minute operation.

3.11.3 Electrical connector. The electrical receptacle on the motor shall be a MS3102-10S-2P with suitable fittings.

3.11.4 Rated voltage. The rated voltage of the unit shall be 115 volts, $\pm$ 10 volts, 400 cycles, single phase.

3.12 Dimensions. The pump unit dimensions, including all projections, shall be as shown on figure 2.

3.13 Weight. The weight of the pump unit, including motor, shall not exceed 5.5 pounds.

3.14 Construction. The pump unit shall be constructed so that no parts will work loose in service. The pump unit shall withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation and service.

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3.14.1 Maintenance. The pump unit shall be constructed so that repairs can be easily made by personnel of operating units and overhaul bases without the use of special tools.

3.15 Performance. The pump units, as specified in paragraph 4.5.2.1, shall satisfactorily complete the first article inspections specified in 4.4.

3.16 Lubrication. The pump unit shall not require external lubrication during its life.

3.17 Threads.

3.17.1 Screw. All machine screw threads shall conform to MIL-S-7742.

3.17.2 Pipe. Pipe threads shall not be used except for permanent closures in which case the pipe threads shall be in accordance with MIL-P-7105.

3.18 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.19 Finish.

3.19.1 Aluminum alloy parts. Aluminum alloy parts shall be anodized in accordance with MIL-A-8625, or chemically treated in accordance with MIL-C-5541.

3.19.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.19.3 Unfinished surfaces. All external surfaces, except anodized alloy, corrosion resisting metals, working surfaces, threads, and plated parts, shall be finished with one coat of primer conforming to TT-P-1757, Type I and one coat of enamel conforming to Specification TT-E-489, Type I, except that the mounting base holes and the bottom of the mounting base shall not be enamelled.

3.20 Marking. The size and location of the inlet and outlet ports shall be marked in the unit and shall be as shown on figure 2. The ports shall be marked "In" and "Out" respectively.

3.21 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The identification data applied to the pump unit shall be as follows:

Pump, Pressure, Oil Free Air, 115 V, 400 Cycle, Single Phase Type MA-3
Specification MIL-P-9389A(USAF)
National Stock Number
Manufacturer's Part Number
Manufacturer'a Serial Number
Contract or Order Number

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Manufacturer's name or trade mark
U.S. Property

3.22 Cleaning. The pump unit shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, or other foreign matter, before and after final assembly.

3.23 Workmanship. The pump unit, including all parts, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of drilling and tapping operation, wiring, soldering, machine screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in this specification.

4.4 First article. First article inspection shall consist of all inspections of section 4.

4.4.1 Calibration inspection. The pump unit shall be placed in an altitude chamber using the apparatus as set up in figure 3. The runs in table I shall be made for voltages of 105 $\pm$ 2 volts, 115 $\pm$ 2 volts, and 125 $\pm$ 2 volts, 400 cycles AC.

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TABLE I. Calibration inspection conditions.

Altitude	Chamber pressure	Inlet pressure P_1	Pump Outlet pressure P_2
Feet	Inches of mercury absolute		
Sea level	29.9 ± 2.0	24.5 ± 0.1	32 ± 0.1
10,000	20.6 ± 0.1	17.6 ± 0.1	32 ± 0.1
20,000	13.75 ± 0.1	11.7 ± 0.1	32 ± 0.1
30,000	8.9 ± 0.1	8.9 ± 0.1	32 ± 0.1
40,000	5.5 ± 0.1	4.8 ± 0.1	32 ± 0.1
50,000	3.4 ± 0.1	3.2 ± 0.1	32 ± 0.1

For each condition specified in table I, record P_o, P_1, P_2, T, T_1, T_2, h_w, and the power watts. The capacity (M), under the conditions specified in figure 3, shall be calculated and plotted on a suitable curve. The chamber temperature during this run shall be $75^{\circ} \pm 15^{\circ}$F. In no case shall the power consumed exceed 750 watts. The pump unit shall operate satisfactorily under all conditions and shall be able to maintain the outlet pressure of 32 inches of mercury absolute throughout the inspection. The capacity shall, in no case, be less than that specified on the curve sheet, figure I.

4.4.2 Low temperature inspection. The pump shall be placed in a cold chamber maintained at a temperature of $65^{\circ} \pm 5^{\circ}$F at sea level ambient pressure, and allowed to soak for 72 hours with test set-up as shown in figure 3. A cold inspection run shall be made for each condition shown in P_1 and P_2 of table I. For each run the pump unit shall be started and allowed to run for 15 minutes, or until terminal temperatures are reached, whichever is sooner, then shut down. The pump unit shall be allowed to cool to chamber temperature and to remain at this temperature for one hour before the next start. This shall be done until all runs in table I are completed. The pump unit shall operate satisfactorily at the completion of the inspection. The pump unit shall not draw more than six amperes at each start, shall draw no more than 4.5 amperes 15 seconds after starting, and after the pump unit has run for one minute the power shall not exceed 350 watts.

4.4.3 High temperature inspection. The pump unit shall be soaked for 24 hours in an ambient temperature of $160^{\circ} \pm 5^{\circ}$F, following which it shall be operated at the following conditions:

Voltage	-	115 ± 2 volts
Frequency	-	400 ± 20 volts
Inlet pressure	-	Ambient room pressure
Outlet pressure	-	32 ± 0.1 inches of mercury

The pump unit shall operate or be operated satisfactorily for 15 minutes with the ambient temperature maintained at $160^{\circ} \pm 5^{\circ}$F.

4.4.4 Environmental inspections. The pump units shall be subjected to the following inspections: Humidity, Sand and dust, Explosion, Fungus, and Salt spray. The Fungus inspection shall be run prior to the Salt spray.

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4.4.4.1 Humidity inspection. The pump shall be placed in the inspection chamber and set up to simulate installed conditions, The inspection chamber shall be vented to the atmosphere to prevent the build up of pressure. Prior to the starting of inspection period, the chamber temperature shall be between 20° and 38°C (68° and 100° F) with uncontrolled humidity. During the first one hour period, the temperature shall be gradually raised to 71°C (160°F). The temperature shall be maintained during the next six hour period. The velocity of the air throughout the inspection area shall not exceed 150 feet per minute. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 20° to 38°C (68° to 100°F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. Steam or distilled water having pH valve between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity. The cycle shall be repeated a sufficient number of times to extend the total time of the inspection to 240 hours (10 cycles). At the conclusion of the 240 hour period, the equipment shall be returned to standard conditions. Moisture shall be removed by turning the equipment upside down or wiping. Drying by air blast or in an oven will be permitted only if so specified. The equipment shall then be operated and inspected within one hour as directed in 4.8.

4.4.4.2 Sand and dust inspection. The equipment shall be placed in the inspection chamber and the sand and dust density raised and maintained at 0.1 to 0.5 grams per cubic foot within inspection space. The inspection chamber shall be vented to the atmosphere. The relative humidity shall not exceed 30 percent at any time. Sand dust used in the inspection shall be of angular structure and shall have characteristics as follows (see 4.11):

a. 100 percent of the sand and dust shall pass through a 100 mesh screen, U.S. Standard Sieve Series.

b. 98 ±2 percent of the sand and dust shall pass through a 140 mesh screen, U.S. Standard Sieve Series.

c. 90 ±2 percent of the sand and dust shall pass through a 200 mesh screen, U.S. Standard Sieve Series.

d. 75 ±2 percent of the sand and dust shall pass through a 325 mesh screen, U.S. Standard Sieve Series.

e. Chemical analysis of the dust shall be as follows:

SUBSTANCE	PERCENT BY WEIGHT
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ign losses	0 to 2

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Part I: The internal temperature of the inspection chamber shall be maintained at 25°C (77°F) for a period of six hours with sand and dust velocity through the inspection chamber between 100 and 500 feet per minute.

Part II: After six hours at the above conditions, the temperature shall be raised to and maintained at 71°C (160°F). These conditions shall be maintained for six hours. At the end this exposure period, the equipment shall be removed and allowed to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, care being taken to avoid introduction of additional dust into the equipment. Under no circumstances shall dust be removed by either blast or vacuum cleaning. The equipment shall be operated and inspected as directed in 4.9.

4.4.4.3 Explosion inspection.

4.4.4.3.1 Fuel. Fuel used shall be gasoline, grade 100/130.

4.4.4.3.2 Purpose. The following procedure shall be used to determine the explosion producing characteristics of items of equipment not equipped with cases designed to prevent flame or explosion propagation.

4.4.4.3.3 Preparation for test.

a. The equipment to be inspected shall be installed in the inspection chamber in such a manner that normal electrical operation is possible and mechanical controls may be operated through the pressure seals from the exterior of the chamber. All external covers of the inspection item shall be removed or opened to insure adequate circulation of the explosive mixture. Two units may be inspected at one time by extending electrical connections through the cable port to the balance of the equipment located externally.

b. The equipment shall be operated to determine that it is functioning properly and to observe the location of any sparking or high temperature components which may constitute potential explosion hazards.

c. Mechanical loads on drive assemblies and servomechanical and electrical loads on switches and relays may be simulated when necessary if proper precaution is given to duplicating the normal load in respect to torque, voltage, current, inductive reactance, etc. In all instances it shall be considered preferable to operate the equipment as it normally functions in the system during service use.

4.4.4.3.4 Inspection procedures.

a. A test shall be conducted as follows:

(1) The inspection chamber shall be sealed and the ambient temperature within shall be raised to +71 $\pm$3°C (+160° $\pm$5°F). The temperature of the inspection item and the chamber walls shall be permitted to rise to within 11°C (20°F) of that of the chamber ambient air, prior to introduction of the explosive mixture.

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(2) The internal inspection chamber pressure shall be reduced sufficiently to simulate an altitude approximately 10,000 feet above the desired inspection altitude. The weight of fuel necessary to produce an air vapor ratio of 13 to 1 at the desired inspection altitude shall be determined from consideration of chamber volume, fuel temperature and specific gravity, chamber air and wall temperature, test altitude, etc. (see 4.7). A time of 3 ±1 minutes shall be allowed for introduction and vaporization of the fuel. Air shall be admitted into the chamber until a simulated altitude of 5,000 feet above the inspection altitude is attained. At this time the potential explosiveness of the resulting air vapor mixture shall be verified by a sampling method which has been approved by the contracting activity.

(3) Operation of inspection item shall at this time be commenced, all making and breaking electrical contacts being actuated. If high temperature components are present, a warm up time of 15 minutes shall be permitted. If no explosion results, air shall be admitted into the chamber so as to steadily reduce the altitude down past the desired test altitude to an elevation 5,000 feet below that altitude or as close thereto as permitted by local ground level altitude. The operation of the test item shall be continuous throughout this period of altitude reduction and all making and breaking electrical contacts shall be operated as frequently as possible.

(4) If by the time the simulated altitude has been reduced to 5,000 feet below the test altitude, no explosion has occured as a result of operation of the equipment under inspection, the potential explosiveness of the air vapor mixture shall again be verified by a sampling method which has been approved by the contracting activity. If the air vapor mixture is not found to be explosive at this time, the inspection shall be considered void and the entire procedure repeated.

b. The above described inspection shall be accomplished at simulated inspection altitudes of local ground level to 5,000 feet, 10,000, 20,000, 30,000, 40,000, and 50,000 feet, unless an explosion is caused by the inspection item, in which case the item shall be considered to have failed to pass the inspection and no further trials need be attempted.

4.4.4.4 Fungus resistance inspection. Four groups of fungi are listed below and one species of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at a temperature between 22°C and 32°C (72°F and 89°F) shall be utilized. Approximately ten milliliters (ml) of the solution shall then be introduced directly into a tube culture of the fungus and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an innoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions from the four types of fungi shall be mixed together to provide a composite suspension. Actively growing cultures between 7 to 21 days old after initial innoculation shall be used for the preparation of the spore suspension.

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After preparation, the spore suspension will not be kept for more than a 24 hour period at temperatures from 22°C to 32°C (72°F to 89°F) or not more than 48 hours at temperatures from 2°C to 7°C (35°F to 45°F). The equipment, including applicable external connections, shall be placed in a mold chamber maintaining an internal temperature of 30° ±2°C (86° ±3.6°F) and relative humidity of 95 ±5 percent, and sprayed with the suspension of mixed spores. The test period shall be 28 days. At the end of the inspection, the parts shall be visually examined in accordance with 4.7. If so specified, the equipment shall be operated and inspected as directed in 4.8.

4.4.4.4.1 Organism.

Group I	-	Chaetomium globosum	6205
		Myrothecium verrucaria	9095
Group II	-	Memenoniella echinata	9597
		Aspergillus niger	6275
Group III	-	Aspergillus flavus	10836
		Aspergillus terreus	10690
Group IV	-	Penicillium citrinum	9849
		Penicillium ochrochloron	9112

4.4.4.4.2 Stock culture designation and source. When ordering cultures, request the cultures by name and serial number. The serial numbers of the cultures remain the same regardless of source.

Source	-	American Type Culture Collection, 2112 M Street, N.W. Washington 6, D. C.

4.4.4.5 Salt spray inspection. The length of the salt spray inspection shall be not less than 168 hours. At the end of the inspection period, the equipment shall be operated and inspected within one hour as directed in 4.8. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water prior to operation. Equipment shall again be operated and inspected 48 hours later, as directed in 4.8.

4.4.4.6 Endurance inspection. After all inspections have been run, the pump unit shall be subjected to the following endurance inspection, at sea level ambient chamber pressure, in accordance with table II. During the inspection, the pump shall cycle 15 minutes running and 45 minutes stopped, except for the final 100 hours of inspection when the cycling period shall be 30 minutes running and 30 minutes stopped. At the completion of the endurance inspection test, the inspection as specified under 4.5.1.2 shall be repeated. The capacity, as recorded following the endurance inspection, shall be not less than 90 percent of the recorded capacities previous to the endurance inspection or the pump shall be rejected. Following these inspections, the pump unit shall be disassembled and inspected for breakage of parts.

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TABLE II. Endurance inspection conditions.

Ambient Temperature	Inlet Pressure	Outlet	Hours	Voltage
(a) $75^{\circ} \pm 15^{\circ}F$	7.0 ± 0.1	32.0 ± 0.1	900	115 ± 2
(b) $75^{\circ} \pm 15^{\circ}F$	28.0 ± 2.0	32.0 ± 0.1	100	115 ± 2

4.5 Quality conformance. Quality conformance inspection shall consist of individual and sampling inspections.

4.5.1 Individual inspections. Each oil free air pressure pump unit shall be subjected to the following inspections:

a. Break-in run
b. Capacity
c. Check valve leakage

4.5.1.1 Break-in run. With inlet and outlet ports open to atmosphere (sea level condition), the pump unit shall be run for 15 minutes at rated voltage.

4.5.1.2 Capacity inspection. The pump unit shall be placed in an altitude chamber using apparatus as shown in figure 3, with P_1 equal to 3.2 ± 0.1 inches of mercury absolute, P_2 at 32 inches ± 0.1 inch of mercury absolute, voltage at 115 ± 2 volts, 400-cycle ac and with inlet and chamber temperature of $75^{\circ} \pm 15^{\circ}F$. The capacity shall be at least 0.00123 lb/min. (30 cu. in per minute at $100^{\circ}F$ and 29.92 inches of mercury absolute.) The chamber pressure shall be 3.40 ±0.1 inches of mercury absolute.

4.5.1.3 Check valve leakage inspection. With the pump unit stationary at sea level, a pressure of 25 ± 0.5 inches of mercury above ambient sea level pressure, shall be applied to the outlet side of the pump and trapped in the air line. There shall be no change in the initial pressure for a period of five minutes.

4.5.2 Sampling inspection.

4.5.2.1 Examination of product. Unless otherwise specified by the contracting activity, oil free air pressure pump units shall be sampled in accordance with the provisions of MIL-STD-105, to determine conformance to this specification with respect to material , workmanship, dimensions, and markings.

4.5.3 Reinspection. Rejected pump units shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection, and measure taken to overcome the defects.

4.6 Air vapor ratio. In illustration of the procecdure for calculating the weight of 100/120 octane gasoline required to produce the desired 13 to 1 air vapor ratio, the following sampling problem is presented:

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squirrel cage rotors shall be used in all motors. 3.2.1 Squirrel cage rotors.

Required information:

a. Chamber air temperature inspection: $80^{\circ}F$.

b. Fuel temperature: $73^{\circ}F$.

c. Specific gravity of fuel at $60^{\circ}F$: 0.704.

d. Inspection altitude: 20,000 feet ($P = 6.75$ lbs/in.^{2}).

e. Air vapor ratio (desired): 13 to 1.

Step 1. Employing the following equation, calculate the apparent air vapor ratio:

$$AAV = \frac{A\text{-}V\ (DESIRED)}{1.04\ \frac{P}{14.696} - .04} = \frac{13}{1.04\ \frac{6.75}{14.696} - .04} = 29.68$$

Where AF = Air Fuel ratio

AAV = Apparent air vapor ratio

AV = Desired air vapor ratio

P = Pressure equivalent of altitude, lbs/in.^{2}

NOTE: At or above 10,000 feet altitude, with chamber air temperature above $60^{\circ}F$ and at AV ratio of 5 or greater, Air vapor ratio = Air fuel ratio for 100/130 octane fuel. Since the conditions of the explosion inspection under consideration will always be well above these values AV will equal AF in all cases.

Step 2. Since AV = AF, use figure 4 to determine weight of air and divide by AAV to obtain uncorrected weight of fuel required.

$$W_{FU} = \frac{Wa}{29.68} = \frac{3.455}{29.68} = 0.116 \text{ lbs, fuel weight (uncorrected)}$$

Step 3. Knowing fuel temperature and specific gravity at 60° F use figure 6 to determine specific gravity at given temperature.

Step 4. Using figure 6 read from specific gravity determined under step 3 for the correction factor k. Apply factor:

$$W_{FC} = W_{FU} \times k = 0.116 \times 1.01 = 0.0117 \text{ lbs, fuel weight (corrected)}$$

NOTE: Figure 5 pertains to a specific test chamber and may not be used for all such test facilities. It is utilized herein for illustration of the method of employment only. Each test chamber must have its own chamber volume chart.

4.7 Vaporizing equipment. The equipment to vaporize the gasoline fuel for use in the explosion proof inspection should be so designed that a small quantity of air and gasoline vapor will be heated together to a temperature such that the gasoline vapor will not condense as it is drawn from the vaporizer into the chamber.

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4.8 Evaluation of equipment. When so directed in individual inspection procedures, the equipment undergoing inspection shall be operated to permit performance data to be obtained, or inspected for evidence of deterioration. The performance data of the equipment under inspection conditions shall be satisfactorily comparable to that obtained in compliance with 4.9. Any deterioration observed shall not exceed that defined in 4.10.

4.9 Performance record. Prior to conducting any of the inspections specified herein, the equipment shall be subjected to a comprehensive operational test under standard conditions and a record made of all data necessary to determine compliance with the specification. These data shall provide the criteria for checking satisfactory performance of the equipment during or after environmental inspection. Where the specification establishes the level of acceptance for the test procedure, a detailed pre-exposure performance record needs to be made. Equipment in this latter category shall be operated prior to test to insure that no malfunction exists.

4.10 Criteria for failure. Deterioration or change in performance of any components which could in any manner prevent the equipment from meeting functional, maintenance and service requirements during service life shall provide reason to consider the equipment as having failed to comply with the conditions of the inspection to which it was subjected.

4.11 Sand and dust. The sand and dust specified for use is commercially known as "140 mesh silica flour". Sand and dust (140 mesh silica flour) produced by the Fenton Foundry Supply Company, Dayton, Ohio, and Ottawa Silica Company, Ottawa, Illinois, or equal, is satisfactory for use in the performance of these inspections.

4.12 Inspection of packaging. Test of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116, to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation packaging. Preservation packaging shall be level A, C, or Industrial as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Pumps shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Pumps shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservation shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each pump shall be individually packaged in accordance with MIL-P-116, submethod II-C insuring compliance with the applicable requirements of that specification.

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5.1.2 Level C. Pumps shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of pumps shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified.

5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A, or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping containers shall be of uniform shape, size, and of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Pumps packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-1672, Type I, Style A. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged pumps shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any special marking required by the contract, each unit packaged and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The oil free air pressure pump unit covered by this specification, is intended for aircraft use in maintaining sea level pressure in systems of low capacity for use in altitudes up to 50,000 feet, or equivalent inlet and surrounding conditions.

6.2 Ordering data.

a. Title, number, and date of this specification

b. Levels of preservation and packing (section 5).

6.3 First Article. When a first article inspection is required, the inspection sample shall consist of two oil free air pressure pump units, selected from the first production lot. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract.

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6.4 Definition of terms. The following definitions shall be applicable to the letters, symbols, etc, as found in figure 3, or elsewhere throughout this specification:

P_o	Absolute pressure in altitude chamber expressed in inches mercury.
P_1	Absolute pressure at pump inlet expressed in inches of mercury.
P_2	Absolute pressure at pump outlet expressed in inches of mercury.
P_3	Barometric pressure at air flow meter expressed in inches of mercury absolute.
h_w	Pressure drop across flow meter expressed in inches of water.
T^w	Temperature of flow meter in degrees Fahrenheit.
T_1	Temperature at pump inlet in degrees Fahrenheit.
T_2	Temperature at pump outlet in degrees Fahrenheit.
M	Rate of flow in pounds per minute
A	Amperes
V	Volts

6.5 Subject term (key word) listing.

Compressor, Air
Compressor, Rotary
Pump, Air, Pressure
Type MA-3

6.6 Changes for previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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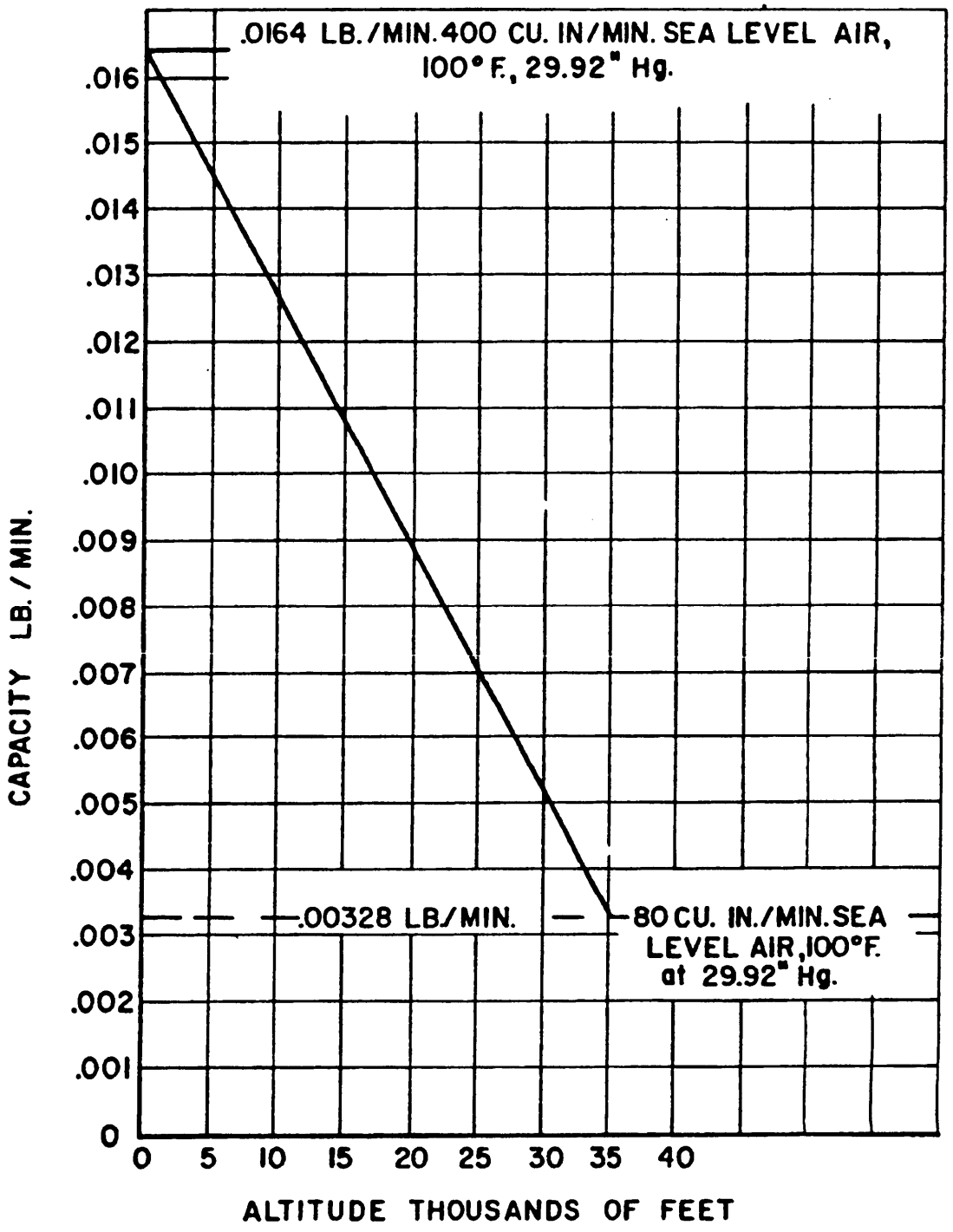


FIGURE 1. Altitude vs capacity.

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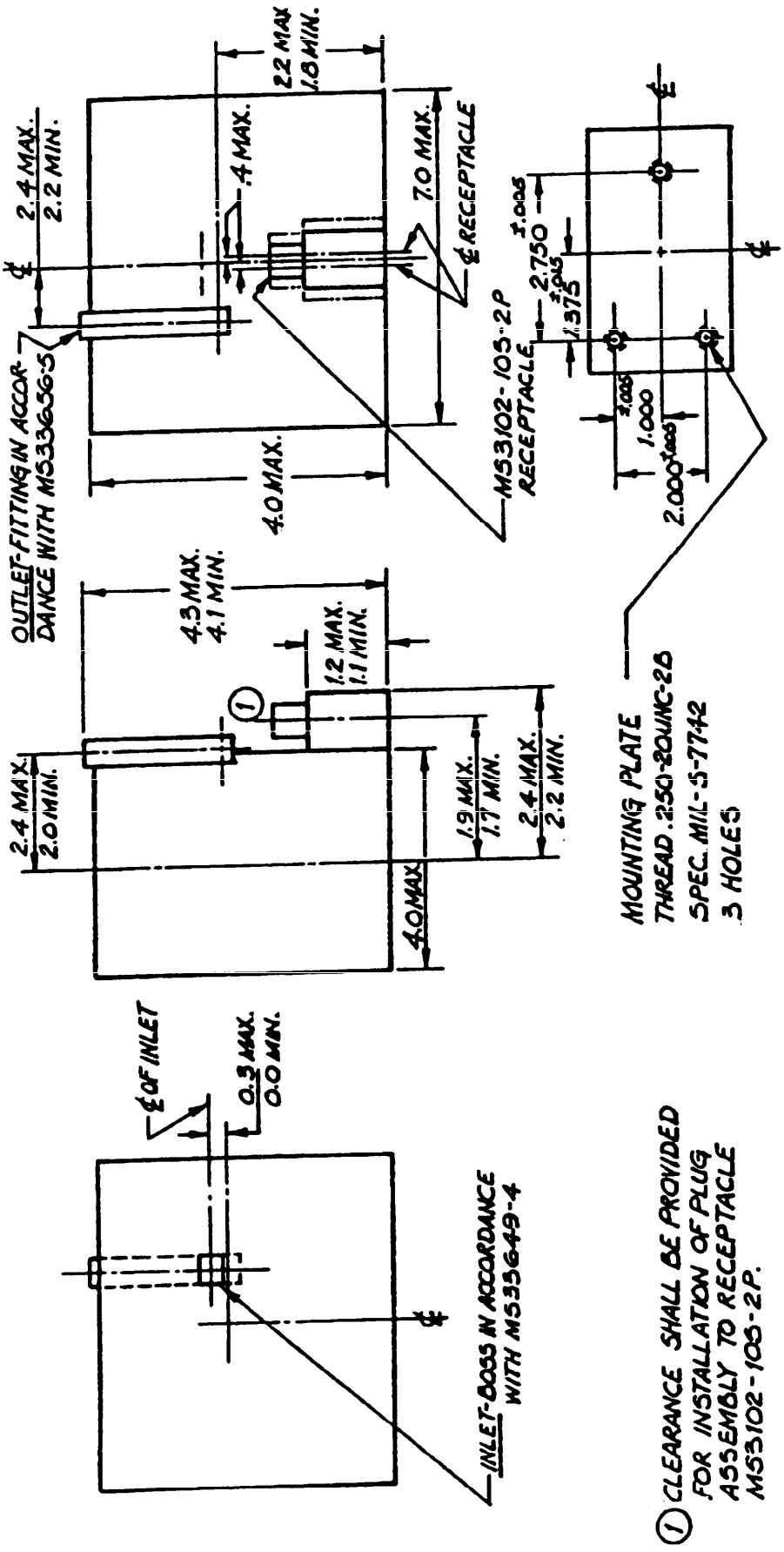


FIGURE 2. Pump dimensions.

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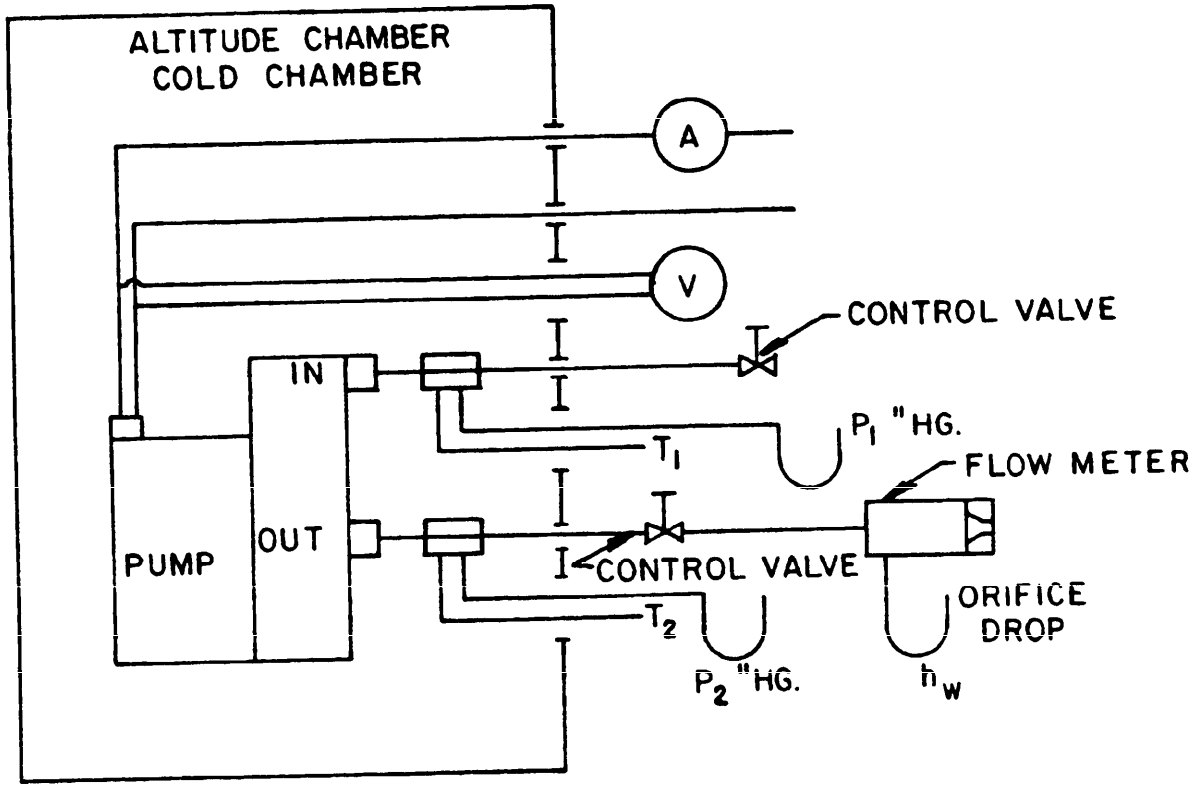


FIGURE 3. Test set-up: calibration, cold test and endurance.

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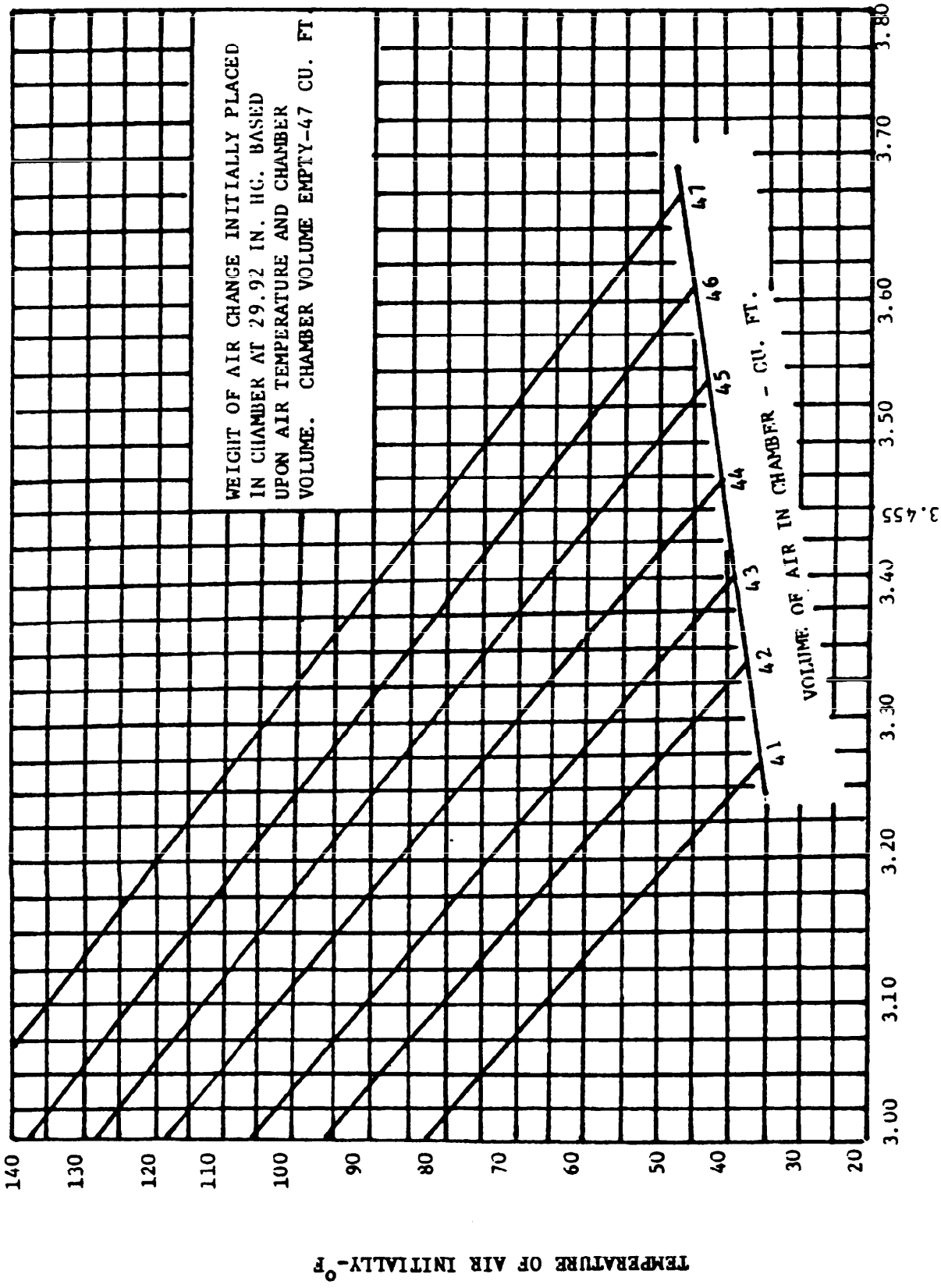


FIGURE 4. Determining weight of air.

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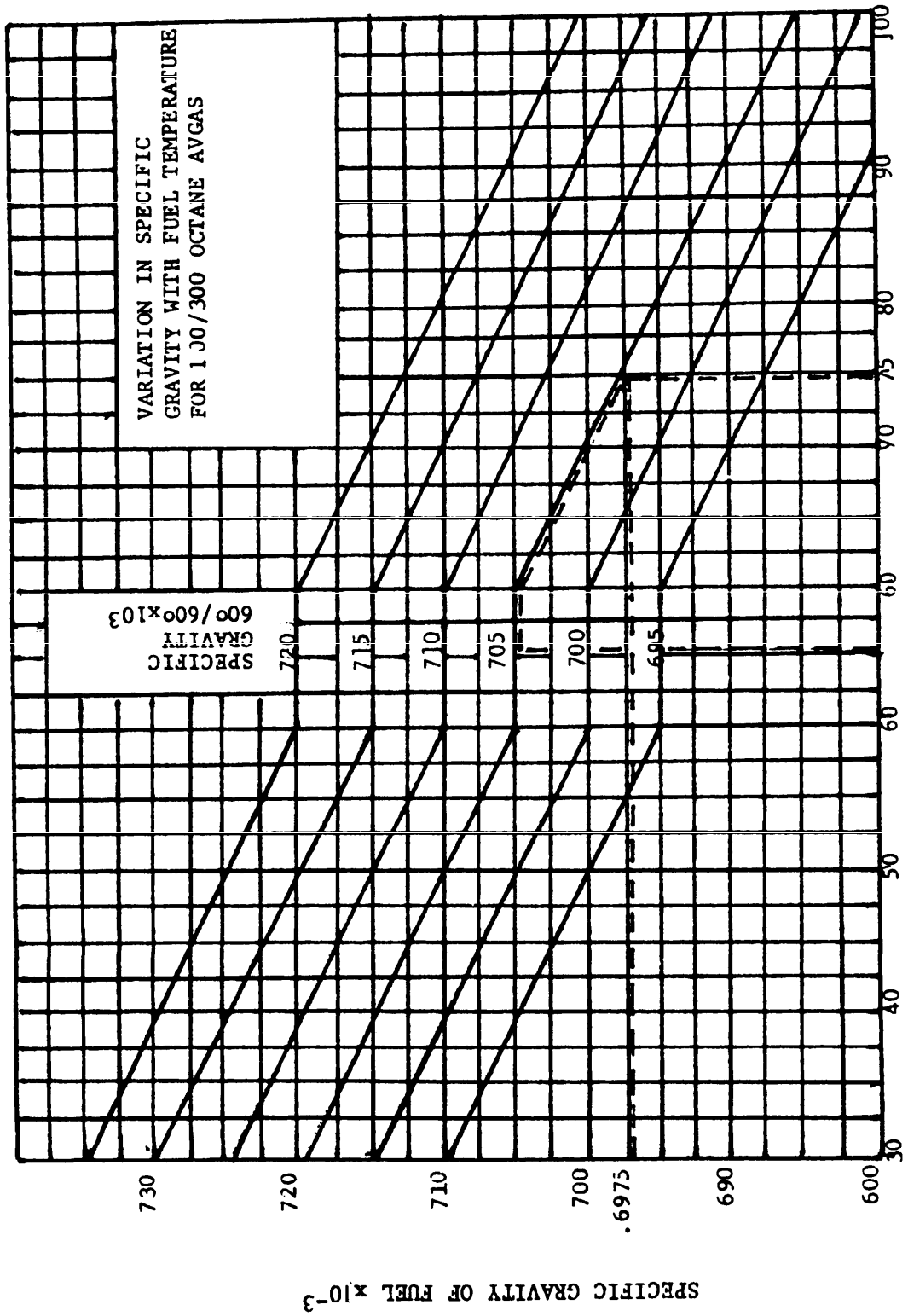


FIGURE 5. Determining specific gravity.

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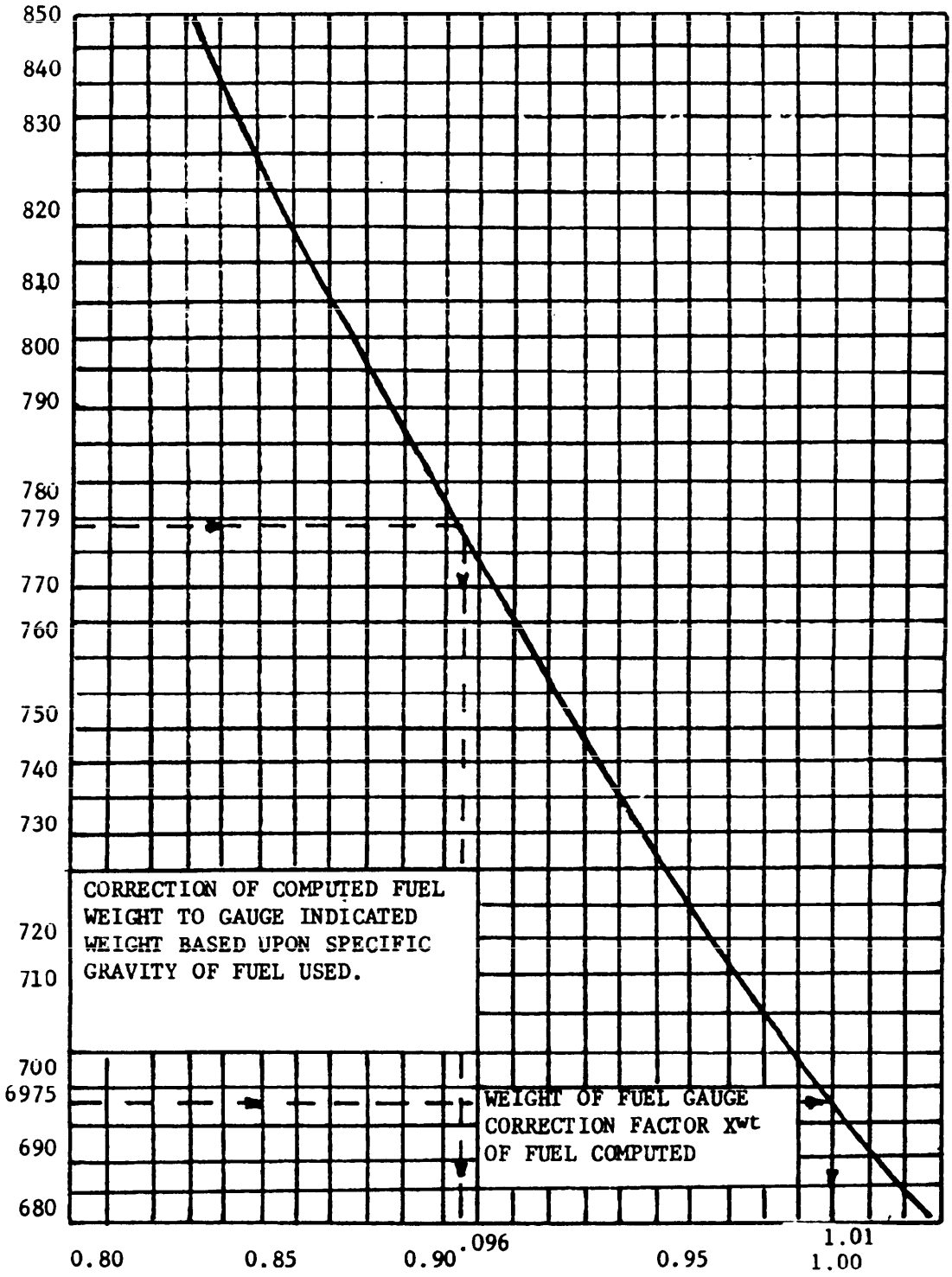


FIGURE 6. Correction Factor vs specific gravity.

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